



UMBC

College of Engineering and Information Technology

Department of Mechanical Engineering



Final Report

Team #12: Morphing Wing

Team Members

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Sponsors

Dr. Meilin Yu, Associate Professor of Mechanical Engineering

Submitted: 5/20/2025

1. Introduction

1.1 Problem Statement

Meilin Yu's research team is focused on developing a morphed wing design in which the leading and trailing edges deform independently. Specifically, the trailing and leading edges must exhibit smooth deformation at 45° and 10° , respectively. The group has also been tasked with using our knowledge of design, coding, manufacturing, control systems, and fluid mechanics to develop both a wireless control system and a smooth surface/covering for the already existing morphed wing design. The smooth outer layer must withstand a variety of atmospheric conditions and be scalable to enable the creation of larger models for use ship-side. Across a series of deliverables, the team will provide research, development, and progress reports to support the final presentation on May 8th, 2026.

1.2 Prior Art

The first prior art was developed by FlexSys, in collaboration with NASA and the Air Force Research Laboratory. The departments created an adaptive trailing-edge control surface under the Adaptive Compliant Trailing Edge (ACTE) program, using FlexFoil. The item serves a similar purpose to the objective of improving a traditional airfoil with a continuously changing shape. It's used for both trailing- and leading-edge deformation, enabling optimized load distribution. The implemented shape-control automation mechanism is useful and can be replicated using a control loop to adjust the trailing-edge shape based on simulated conditions. The product has also been flight tested, demonstrating a 20% reduction in actuation power compared to traditional flaps. [1]

The second prior art is the Fishbone Active Camber Morphing Wing, devised by Benjamin K.S. Woods and Michael I. Friswell at the University of Bristol (UK). It is a morphing wing inspired by fish fins that focuses on high-bandwidth, continuous deformation. Similar to the team's morphing wing, the airfoil provides an alternative design architecture to traditional flaps. This design is unique because structural deformation occurs through compliance, without mechanisms or linkages. The Elastomeric material has demonstrated aerodynamic performance exceeding that of conventional flaps while allowing independent control of the trailing edge, which aligns with the project's design parameters. [2]

Droop Nose Leading Edge is an advanced, smart-skin, wing-shaping technology that enhances aerodynamic performance by eliminating traditional discrete gaps and hinges. This functions similarly to the design we aim to achieve. The seamless trailing edge changes camber without gaps, reducing the pressure drag compared to conventional hinged flaps. Applied to the UAS-S45 airfoil, there was a 12.18% drag reduction and an increase of up to 10% aerodynamic endurance. These results were observed using a novel algorithm; thus, similar observations will be harder to obtain. [3]

The belt-rib concept for variable-camber airfoils was developed at the German Aerospace Center as part of an Adaptive Wing project. The concept features a flexible skin attached to a closed-loop belt that forms the airfoil shape. It is designed to maintain structural integrity against aerodynamic loads while allowing elastic deformation. The shape change is driven by actuators, which operate differently from those in a standalone wireless system. However, the optional physics-informed system will be comparable but will not include materials such as shape memory alloys. [4]

Another new morphing wing technology was developed under the U.S. Air Force Research Laboratory's patent for a 'Morphing Airfoil'. The system features a dynamic, flexible skin with actuated

ribs and a built-in control framework that allows wireless control. The team will study their research to gain insight into how the wireless system deforms the trailing and leading edges independently.[5]

2. Preliminary Design Work

2.1 Customer Needs

The morphing wing sponsor, Dr. Meilin has requested smoother deformation of the leading and trailing edges, a new wireless control system, a smooth skin covering the wing, and a design that is scalable and easily manufactured. Yu requires the deformation of the leading and trailing edges to be 45° and 10° , respectively.

2.2 Mission Scenario Diagram

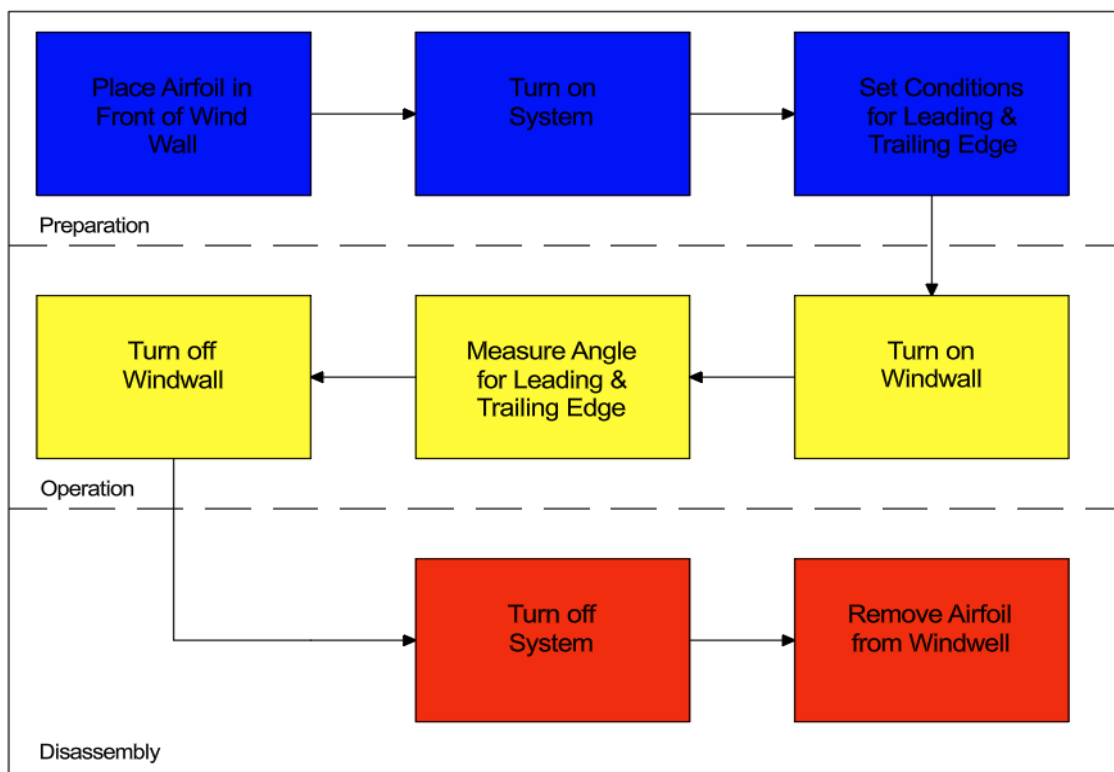


Figure 1: Mission Scenario Diagram

The Mission Scenario Diagram (MSD) displays the three-phase process for properly deploying the morphing wing prototype for testing in a wind tunnel. The modifications to wireless controls and a smoother wing skin will not affect the deployment process from the previous capstone team. To properly deploy the prototype, it is carefully placed in the windwell, turned on, and the desired leading- and trailing-edge conditions are set wirelessly. During testing, the angles should be measured, and the windwell turned on and off. After testing is complete, the morphing wing controls and power should be properly shut down, then carefully transported and stored.

2.3 Black Box Diagram

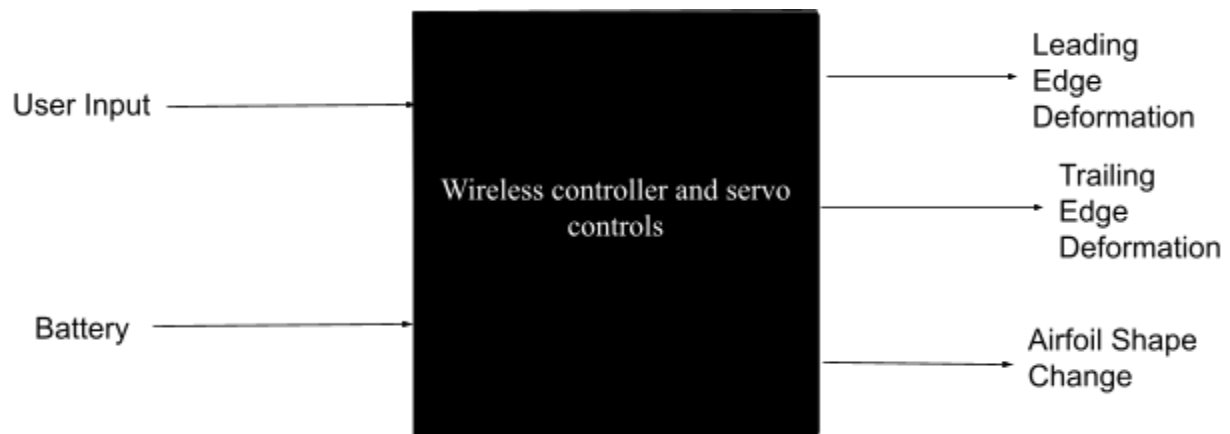


Figure 2: Black Box Diagram

Figure 2.3, the black-box diagram, shows that power from the battery and user input enter the system, while leading-edge and trailing-edge deformation are the outputs.

2.4 Function Structure Diagram

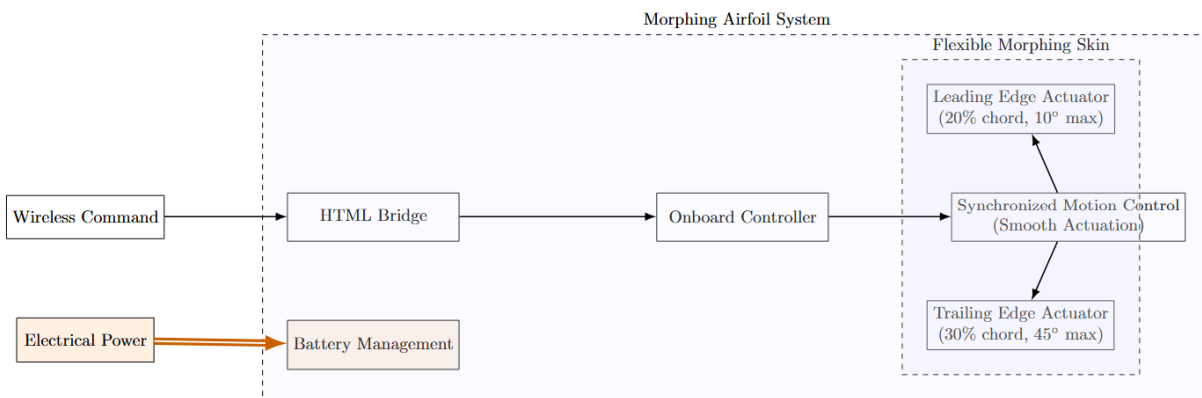


Figure 3: Function Structure Diagram

The function diagram depicts the subsystems necessary for device operation. Arrows represent the relations between the subsystems. The wireless command to the HTML bridge stream flow represents the morphing wing's operational process. Once the trailing edge receives information from the controller, it will apply the desired deformation to improve aerodynamic performance. The synchronized motion control ensures the airfoil can pitch both edges simultaneously with smooth motion. The highlighted section is the power distribution flow for the system.

2.5 Specifications

		Metrics			
ID	Customer Needs	Thickness	Cost vs. Wired Control	Number of Sensors	Angle of curvature
1	Covering that provides "Smooth Flow."	x			x
2	Wireless Control of Bending		x		
3	(Optional) Feedback loop sensor			x	
4	Deformation of Leading and Trailing Edges				x
	Target Value	1/16"	> \$150	4	Leading Edge: 10 Trailing Edge: 45
	Maximum Val.	1/8"	\$200	6	Leading Edge: 12 Trailing Edge: 43
	Minimum Val.	1/32"	\$90	2	Leading Edge: 9 Trailing Edge: 44

Figure 4: Needs Metrics Matrix

Figure 4 shows the customer needs and their corresponding target metrics, which coalesce into the specifications sought. The columns represent the quantifiable metrics used to measure whether these specifications have been satisfied, and the rows represent the customer needs being addressed.

ID	Customer Needs	Rationale	Verification
1	Covering that provides "Smooth Flow."	Sponsor Request	Performance Evaluation
2	Wireless Control of Bending	Benchmarking	Performance Evaluation
3	(Optional) Feedback loop sensor	Benchmarking	Performance Evaluation
4	Deformation of Leading and Trailing Edges	Sponsor Request	Theory

Figure 5: Rationale and Verification Methods

Figure 5 presents the rationale for the chosen metrics for each customer need, and the verification method used to measure each specification.

3. Concept Development

3.1 Subsystem Design Concepts

Group 12 has zeroed in on the desired subsystems, ordered, and received them in the mail. Since the project is geared toward design agendas and the integration of new systems into an existing system, the focus was on understanding how the current wired control systems interface with the servo motors on the back and front ends. Analyzing the code, understanding how the controller is mounted, and even studying wiring schematics, the focus is on using current integration methods to add a wireless Wi-Fi controller. As seen in Figures 9 and 10, the concept choices were narrowed down using the 3-6-5 concept generation technique. Furthermore, all initial concepts for each subsystem are compared using their pros and cons. Subsystems are divided into two categories: wireless connectivity and control, and smooth aerodynamically focused coating for the airfoil frame.

3.2 Concept Screening

	Concept Variants			
Selection Criteria	Arduino UNO Q 4-GB RAM	Arduino GIGA R1 WIFI	Neoprene Rubber sheet	Industrial Latex
Smooth Airfoil covering	0	0	+	+
Wireless Control of Bending	+	+	0	0
Feedback loop sensor	+	-	0	0
Deformation of lead/trailing edges	+	+	+	+
Cost	+	+	+	-
Ease of Implementation	+	-	+	-
Pluses	5	3	4	2
Sames	1	1	2	2
Minuses	0	2	0	2
Net	5	1	4	0
Rank	1A	2A	1B	2B
Continue?	Yes	No	Yes	No

Figure 6: Concept Screening Matrix

Figure 6 shows a concept screening matrix to evaluate top-performing candidate subsystems. Concept variants in the screening matrix include Arduino UNO Q with 4 GB RAM, Arduino GIGA R1 WIFI, Neoprene Rubber sheet, and Industrial Latex. For selection criteria, the screening matrix consists of Smooth Airfoil covering, Wireless Control of Bending, Feedback loop sensor, Deformation of lead/trailing edges, Cost, and Ease of Implantation. The top two ranking concept variations are Arduino UNO Q with 4 GB RAM for wireless control and Neoprene Rubber sheet for a smooth airfoil covering.

3.3 Concept Scoring

		Neoprene Rubber	Industrial Latex
Criteria	Weight	Score	Score
Flexibility	30%	0.35	0.65
Durability	25%	0.70	0.30
Temperature Resistance	20%	0.80	0.20
Scalability	15%	0.65	0.35
Cost	10%	0.30	0.70
Total Score		0.57	0.43
Continue?		Yes	No

Figure 7: Concept Scoring for Smooth Skin

		Arduino UNO R4	Arduino GIGA R1
Criteria	Weight	Score	Score
Capability	30%	0.30	0.70
Ease of Implementation	25%	0.70	0.30
Cost	20%	0.75	0.25
Connectivity	15%	0.50	0.50
Thermal Resistance	10%	0.60	0.40
Total Score		0.58	0.42
Continue?		Yes	No

Figure 8: Concept Scoring for Wireless Control

Neoprene Rubber and Arduino UNO R4 were the two concepts selected for the product. Neoprene was selected because it aligned best with the project's requirements for a viable subsystem. While the Neoprene did not score highest in the weightiest criterion (flexibility), it scored higher than industrial latex in scalability, durability, and temperature resistance. Factoring in industrial use, this

conclusion is consistent with the goal of widespread application. The rubber will withstand adverse temperatures common in the aircraft and aerodynamic industries.

Additionally, in an industry that features wings of various sizes, the material can be used in smaller scales or fit larger aircraft, such as planes or larger UAVs. For the control system, the Arduino UNO R4 was the superior choice due to ease of implementation and cost. In this case, and in many industrial applications, control systems are built by integrating multiple subsystems. Thus, using the scoring matrix, it was determined that the Arduino UNO R4 would be selected because of its portability with the existing Arduino Mega and other various software. Proceeding with this, the Arduino R4 also made sense because it's cheaper, considering the possibility of additional expenses for this project or in the engineering industry.

3.4 Concept Selection

Concept Name	Concept Description	Pros	Const
Neoprene Rubber sheet coating fastened onto the main frame of the airfoil, as well as the leading and trailing edges	Thin synthetic rubber that can act as a covering for the airfoil. Greater focus on industry application and scalability.	● Oil/heat-resistant rubber ● Durable. Ideal for high-temperature applications. ● Tear-resistant.	● High Cost ● Non Biodegradable
Industrial Latex covering is used mainly to cover the frame of the airfoil	Natural rubber is used to coat the airfoil. An adequate representation of other, more industrially used latex materials when scaled down as a model.	● Flexible ● Highly Elastic ● Natural Rubber	● Sensitive to temperature change ● It would impede the deformation of the trailing edge due to its thickness.

Figure 9: Concept selection for Smooth Skin

Concept Name	Concept Description	Pros	Cons
Arduino UNO Q 4-GB RAM wireless control integration with existing controller	Wireless development board with Wi-Fi connectivity and an integrated microcontroller that interfaces with an existing controller to enable wireless operation.	• 4GB RAM • 32GB Storage • Wifi/Bluetooth connection • Low cost	• The device may run hot. • Cheaper options with similar capabilities
Arduino GIGA R1 WIFI standalone control of the deformation	A powerful microcontroller handles all the heavy lifting of controlling the system and uses Wi-Fi for remote interfacing.	• A dual-core processor for multitasking • Supports 24V power input	• Significantly higher cost • Overly Complex (Would require a level of expertise to work with)

Figure 10: Concept Selection for Wireless Controls

4. Prototyping

4.1 Initial Prototypes

The initial prototype features hinges that maintain a smooth, continuous curvature. The group has defined “smooth” as the trailing edge parts remaining uniform during angle deformation. At the trailing edge, the subsystem sustained 120 degrees of motion. The leading edge could sustain 20 degrees of motion. For both the trailing edge and leading edge, there was no stress whitening after repeated actuation. Regarding any subsystem problems, two servos on the trailing edge were found to be defunct after testing. The trailing and leading edges were functional after the two servos were replaced.

The wireless control works through established network communication between the Arduino Uno and the Linux laptop software via Secure Shell (SSH) over Wi-Fi. This process allows the Arduino to receive and execute commands. The issues encountered were related to troubleshooting the correct interfaces for the connection between the laptop and the Arduino and streamlining the command-sending process. After realizing that the official Arduino lab app could not connect to networks outside the restricted environment, it was discovered that the Arduino could communicate with the laptop via Linux. Once the network was established, debugging consisted of determining the angles the servos could receive.

The trailing edge subsystem, comprising the pin and hinge system, meets the 45-degree angle deformation criteria while maintaining a single body throughout movement. After approximately 150 angle commands, no visual defects or whitening are observed on the body. The pins are still intact and firmly in place at the trailing edge subsystem. The leading-edge subsystem also meets the design requirements, deforming by 10 degrees independently.

The wireless control system functions as described in previous memos, with the laptop sending commands to the morphing wing remotely. The Arduino Uno serves as both a receiver and a controller, receiving commands from the network and simultaneously controlling the servos. The subsystem's

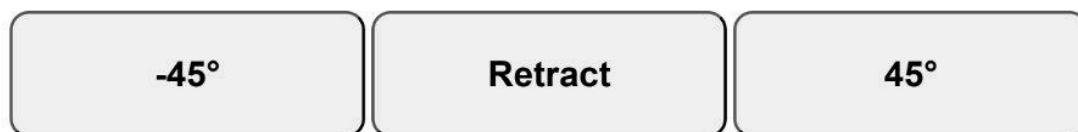
simplicity allows future groups to wire more components to the Arduino Uno and provides them with more room within the airfoil.

4.2 Final Prototype

The final prototype includes a wireless control system with correctly mapped inputs and outputs. In addition, a user interface was created for a more intuitive control apparatus for those unfamiliar with the Python-Linux command environment.

Wing Controls

Trailing Edge



Leading Edge



READY

Figure 11: New user interface for intuitive control

Within the leading and trailing edge subsystem, new seating consoles for each servo act as servo mounts, stabilizing the movement of the lead and tail deformation. A bar-and-pin configuration connects the trailing edge to the main body, allowing smooth axial deformation at specified angles in the trailing edge.

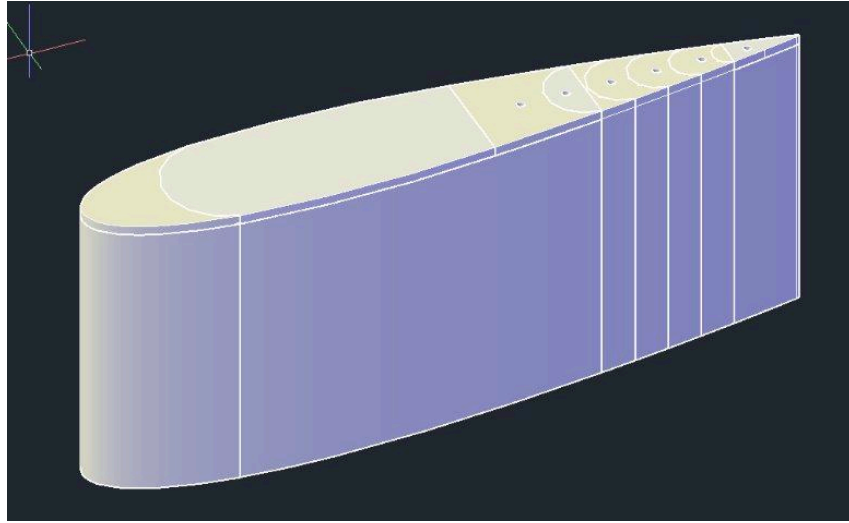


Figure 12: Holes for bar and pin configuration implemented in the design

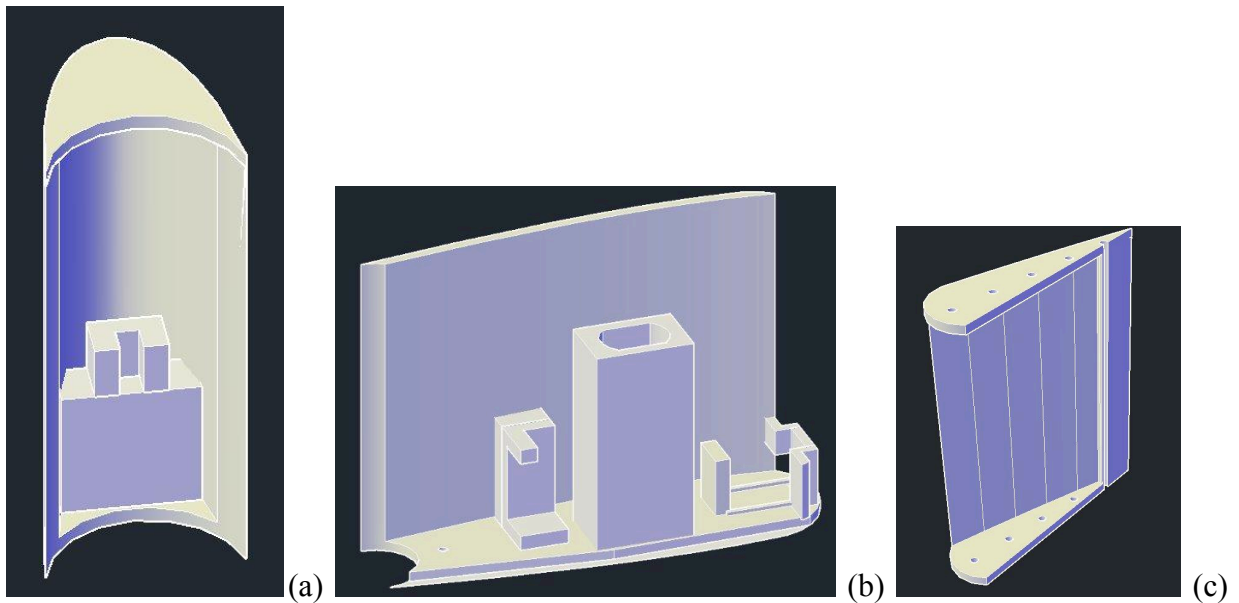


Figure 13: Seating Consoles acting as servo mounts for each deforming edge. Leading edge (a) main body (b) trailing edge (c)

Additionally, a neoprene rubber sheet has been fitted to the outer surface of the airfoil compartments. This covering has been implemented specifically for the leading edge, the terminal scale of the trailing edge, and the main body. The rubber sheets are arranged in an overlapping configuration to allow free, independent movement of the three main subcompartments of the airfoil.

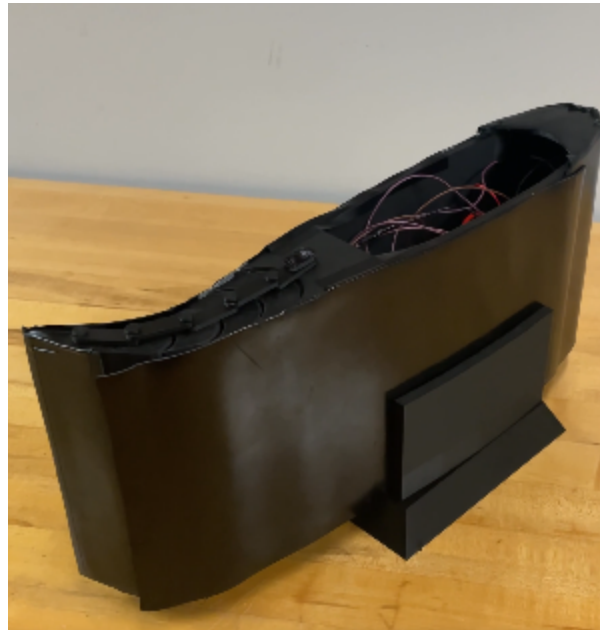


Figure 14: Neoprene rubber sheet configuration as the trailing edge deforms

5. Testing and Verification

5.1 Rationale and Verification Methods

Requirement	Rationale	Verification
Wireless	Remote morphing capability	Test using a laptop/phone
Lightweight construction	Prototype feasibility	Material selection
Leading Edge deformation	Small controllable deformation	Actuation testing
Trailing edge deformation	Large controllable deformation	Actuation testing
10° deformation	Requirements	Measured angle after deformation
45° deformation	Requirements	Measured angle after deformation
Smooth servo motion	Remove twitching of servos	Observable during testing
Repeated deformation cycles	Consistent performance	Multicycle testing

Smooth Outercovering	Surface deformation	Visual inspection
Electronics	Future scalability	Subsystem testing

Figure 15: Rationale and Verification Methods

All our verification processes relied on practical tests conducted on the actual model we designed. The main aim of this whole exercise is to prove the feasibility of this technology. So, much attention was paid to morphing functionality, wireless control, and overall structural behavior rather than aerodynamic efficiency optimization. The deformation angles for both the leading and trailing edges were determined in accordance with the project's initial criteria.

5.2 Verification

Requirement	Quantity Measured	Verification	Result	Status
Wireless	Responsiveness and range	Testing using a laptop/phone	Stable wireless control	Achieved
Leading-edge deformation	Angular Displacement	Actuation Testing	10° deformation	Achieved
Trailing edge deformation	Angular Displacement	Actuation Testing	45° deformation	Achieved
Smooth Outercovering	Surface Roughness	Visual inspection	Continuous outer profile during deformation	Achieved

Figure 16: Verification Table

Servo actuation testing was performed to verify that the airfoil achieved the required morphing angles. The leading edge was required to deform approximately 10°, while the trailing edge was required to deform approximately 45°. Angular displacement was measured visually relative to the undeformed airfoil configuration during operation. The servos consistently achieved the target deformation ranges without stalling or excessive vibration.

Wireless Control Test 1 Trailing Edge

Angle of Servo Input	Wing displacement angle
100	50
55	-10
10	-45
55	10
100	60
55	-10

Wireless Control Test 2 Trailing Edge

Angle of Servo Input	Wing displacement angle
90	50
45	-20
10	-45
50	-5
52	-10
90	50

Figure 17: Mapping from input within the command shell to the physical angle of deformation for the trailing edge

Wireless Control Test Leading Edge

Angle of Servo Input	Wing displacement angle
120	0
130	10
105	-10

Figure 18: Mapping from input within the command shell to the physical angle of deformation for the leading edge

In Figure 18, we performed only one test for the leading edge because it was directly connected. In Figure 17, we conducted numerous tests on the trailing edge due to a gap between the bent rod and the linkage. Every time we rotated the bent rod, the servo input angle changed.

6. Reflection and Future Work

6.1 Reflection

For the trailing edge, the team modified the existing design for holding the segments together with fishing wire to a bar-and-pin system. Originally, it was decided to use metal pins with 3D-printed bars, but it was realized that the way the pins were secured led to significant side-to-side movement. The metal pins were secured with cotter pins inserted into holes near the bottom of the pins, leaving a quarter-inch gap. A solution was to use 3D-printed pins, which were melted into the inside of the segments once the proper staggered bar formation was established. This led to the bars being the only moving parts on the trailing edge, allowing smooth motion without side-to-side motion. The only downside to this method was that the last connection on the body had to have the 3D-printed pins facing outward and be melted down on a half-cut mounting bar, since melting the pins on the moving bar would reduce mobility. Another downside of this mounting method is that when the trailing edge needs to be removed, the mounting pins must be snipped, and the remounting process takes 20 minutes.

Originally, the leading edge seemed to work, but the mount needed to be bigger with side walls to stop the bar from breaking off the mount. A new servo mount console was developed. Heat-set inserts were added so that fasteners could be used to attach the servo to the leading edge, which would propagate the deformation. Fasteners that matched the threading of the inserts were used to avoid cross-threading or stripping of critical screws during construction or repair.

The wireless control system was implemented as intended. When users selected a degree angle on the HTML site, the morphing wing achieved the deflection parameter with minimal lag. For ease of use, the website is configured to launch whenever the Arduino is powered. Future teams can enhance the subsystem by seamlessly editing the HTML code. When verifying the deflection angles, users had to log in to the Arduino HTML folder and refresh the page for the changes to take effect. Another potential solution for the subsystem would be to assign a permanent location for the control site. Due to the dynamic IP address, users may need to connect to the board via the Arduino Lab App to find the HTML site's IP address.

Although the outer neoprene rubber coating provided a smooth surface for an aerodynamic profile during morphing, wrinkles were observed as deflections increased at the trailing edges. This shows that improvements can be made regarding the thickness and bonding method for attaching the skin. Future projects will look into introducing the skin thickness and possibly even using a composite. Stretchable composites can stretch better but maintain smooth aerodynamics. Another area that will receive further attention is finding the optimal method of attaching the skin to the internal structure.

6.2 Future Work

The wireless control system, though robust, is still manually operated by the user. While this is sufficient for the current project, a closed-loop feedback control system that enables autonomous course

correction, stabilization, and feedback response data would be useful for future research on morphed wing interactions under various fluid flow conditions.

Due to repeated loosening and tightening, the collar that secures the precisely bent rod to the servo in the trailing edge has stripped Allen key screws, and the screw that holds the rod cannot be tightened or removed. Designing or purchasing a more secure and robust shaft collar system might be beneficial for subsequent prototyping.

Finally, as noted in the reflection, 3D-printed pins worked with the trailing edge segments, except for the section that interfaces with the main body. A more secure pin-mount system can be designed for this specific section of the trailing edge.

7. References

- Include all references (patents, standards, personal conversations, etc.) in IEEE format.

[1] S. Kota and J. R. R. A. Martins, “FlexFoil Shape Adaptive Control Surfaces—Flight Test and Numerical Results,” in *Proceedings of the 31st Congress of the International Council of the Aeronautical Sciences (ICAS 2018)*, Belo Horizonte, Brazil, Sep. 9–14, 2018. :contentReference[oaicite:0]{index=0}

[2] B. K. S. Woods, J. H. S. Fincham, and M. I. Friswell, “Aerodynamic Modeling of the Fish Bone Active Camber Morphing Concept,” presented at *RAeS Applied Aerodynamics Conference*, Bristol, U.K., Jul. 22, 2014. :contentReference[oaicite:2]{index=2}

[3] Bashir, M., Longtin-Martel, S., Botez, R. M., and Wong, T., “New Wing Morphing Technologies in Support of Next Generation Drones,” ÉTS Montréal News, Oct. 26, 2022.

[4] Campanile, L. F. and Anders, S., “Aerodynamic and Aeroelastic Amplification in Adaptive Belt-Rib Airfoils,” *Aerospace Science and Technology*, Vol. 9, No. 1, Jan. 2005, pp. 55–63. doi: 10.1016/j.ast.2004.07.007

[5] Moudgal, S., & Sharma, M. (2025). “*Systems and methods for a conversational agent using a large language model with contextual augmentation*” (U.S. Patent No. US 12,163,444 B1). U.S. Patent and Trademark Office. <https://patents.google.com/patent/US12163444B1/en>

APPENDIX A: Work Breakdown Structure and Gantt Chart

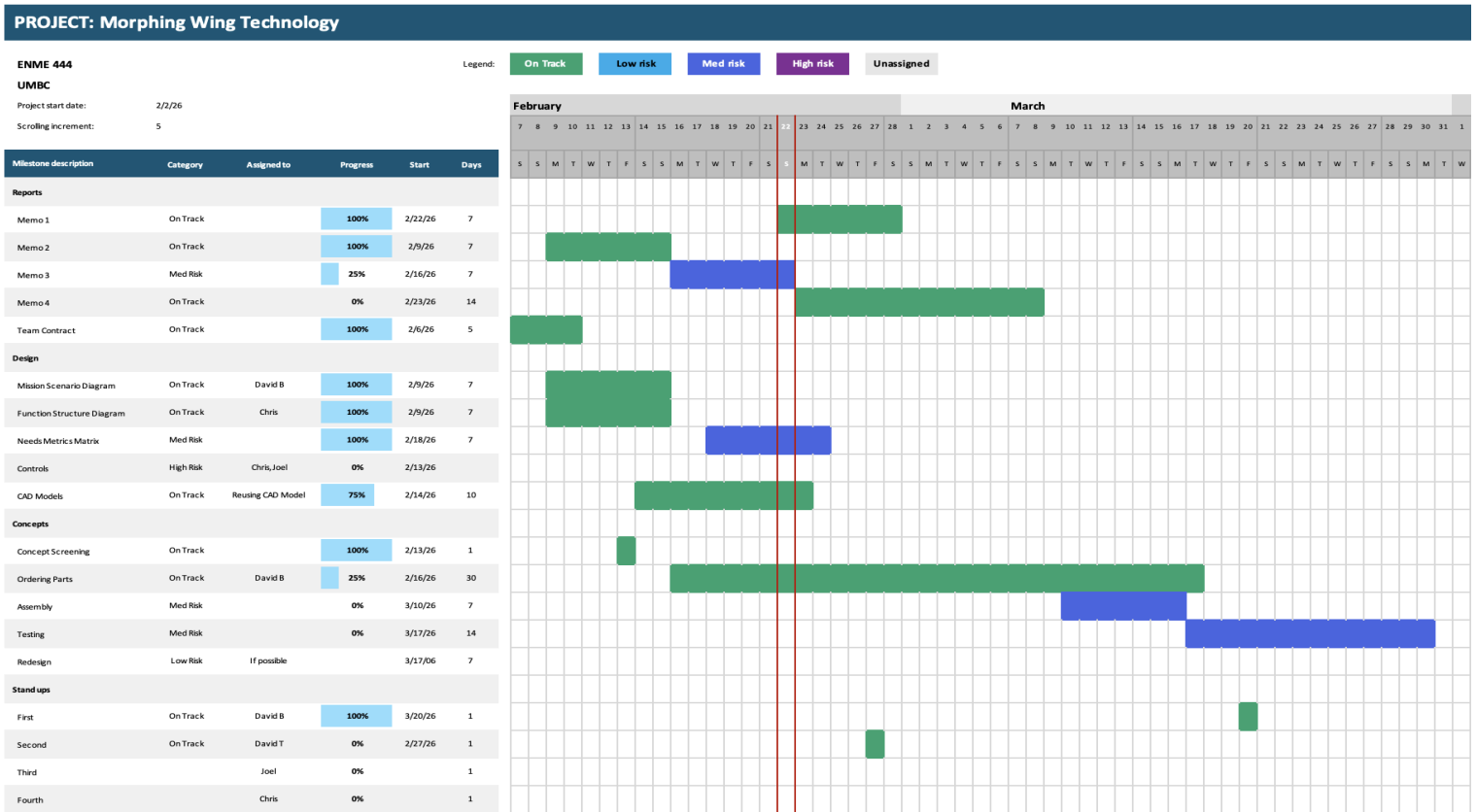


Figure 19: Gantt Chart

Work Breakdown Structure

The Product Research Activity consists of the following work packages: creating a problem statement, identifying customer needs, and researching existing technology. The customer needs identified were upgrading to wireless controls, creating a smooth, flexible wing skin, finding industrial applications, and improving interaction between the trailing edge and PBR for smoother deformation. The concept development activity consists of creating wireless controls using an Arduino Nano ESP32, creating a smooth wing skin with a neoprene sheet, and developing a mechanism to improve interaction between the trailing edge and PBR by modifying the existing CAD models. The concept prototyping package consists of testing and calibrating the wireless controls and developing a frame to hold the neoprene rubber onto the wing. When testing the wireless controls, it must be confirmed that the leading edge deforms 20% of chord length with a max deform angle of 10°, and the trailing edge deforms 30% of chord length with a max deform angle of 45°. The documentation work package consists of our team contract, project memos, individual standups, project presentations, and our final project documentation.

1. Product Research (Activity)

- a. Create Problem statement (**work package**)
 - i. Upgrading the morphing wing prototype from the previous semester (**tasks**)
- b. Identify customers' needs (**work package**)
 - i. Upgraded wireless controls (**tasks**) [Chris]
 - ii. Smoother wing cover (**tasks**) [David B]
 - iii. Industrial Applications (Optional) (**tasks**) [Chris]
 - iv. Better interaction between the trailing edge and PBR, giving a smoother deformation (Optional) (**tasks**) [David I]
- c. Researching existing technology (**work package**)
 - i. Morphing wing technology (**tasks**) [David B]
 - ii. Aerospace-grade latex (**tasks**) [David B]
 - iii. Polymer-based skins (**tasks**) [David B]
 - 1. Electroactive polymers
 - 2. Shape-memory polymers
 - 3. Polymetric skins

2. Concept Development (Activity)

- a. Creating wireless controls for the existing wing (**work package**)
 - i. Get existing code from the previous capstone team and reprogram for wireless controls (**tasks**) [Chris]
 - ii. Order Arduino Nano ESP32 (**tasks**) [David B]
- b. Creating a smooth wing (**work package**)
 - i. Order Neoprene Rubber Sheet - 1/16" Thick (**tasks**) [David B]
- c. Modify the mechanism for better interaction between the trailing edge and PBR(**work package**) [Joel]

3. Concept Prototyping (Activity)

- a. Testing wireless controls (**work package**)
 - i. Leading Edge - deform 20% of chord length, max deform angle of 10° (**tasks**) [Chris]
 - ii. Trailing Edge - Deform 30% of chord length, max deform angle of 45° (**tasks**) [Chris]
 - iii. Rewire prototype and test new control system(**tasks**) [Joel]
- b. Add smooth, flexible skin
 - i. Create a frame to mount Neoprene Rubber Sheet (**tasks**) [David B]
 - ii. Test skin flexibility with the Leading/Trailing edge at maximum angle (**tasks**) [David B]
 - iii. Test aerodynamics in wind tunnel (**tasks**) [Joel]
 - 1. Compare performance with the previous team's results

4. Documentation (Activity)

- a. Team Contract (**work package**)
- b. Project Memos (**work package**)

- i. Memo 1: Problem Statement, Prior art (**work package**) [**All members**]
 - ii. Memo 2: Customer needs, Mission Scenario Diagram, and Functional Block Diagram (**work package**) [**All members**]
 - iii. Memo 3: Specifications, Work Breakdown Structure, and Gantt Chart (**work package**) [**All members**]
 - iv. Memo 4: Design Concept Evaluation and Selection (**work package**) [**All members**]
- c. Standups
 - i. Standup 1 (**tasks**) [**David B**]
 - ii. Standup 2 (**tasks**) [**David I**]
 - iii. Standup 3 (**tasks**) [**Chris**]
 - iv. Standup 4 (**tasks**) [**Joel**]
- d. Project Presentations
 - i. Project Presentation 1: Design Concept Evaluation and Selection (**work package**) [**All members**]
 - ii. Project Presentation 2: Alpha 1 Prototype (**work package**) [**All members**]
 - iii. Project Presentation 3: Alpha 2 Prototype (**work package**) [**All members**]
 - iv. Project Presentation 4: Beta 1 Prototype / Preliminary Design Review (**work package**) [**All members**]
- e. Final Project Documentation
 - i. Capstone Design Presentation (**work package**) [**All members**]
 - ii. Final Project Presentation (**work package**) [**All members**]
 - iii. 30 Minute Project Demo (**work package**) [**All members**]
 - iv. Final project Report (**work package**) [**All members**]

APPENDIX B: Initial Subsystem Concepts (include if needed)

- This section can draw from the Subsystem Concepts section of Memo 4
- If not used, place “This Page Intentionally left blank” beneath the “Appendix B: Initial ...” header for the page

APPENDIX C: Analytical Hierarchy Process (AHP)

	Flexibility	Durability	Temperature Resistance	Scalability	Cost	Total	Weight
Flexibility		1.2	1.5	2	3	7.7	0.3
Durability	0.83		1.25	1.67	2.5	6.25	0.25
Temperature Resistance	0.67	0.8		1.33	2	4.8	0.2
Scalability	0.5	0.6	0.75		1.5	3.35	0.15
Cost	0.33	0.4	0.5	0.67		1.9	0.1
Total						24	1

Figure 20: AHP - Smooth Cover

	Capability	Ease of Implementation	Cost	Connectivity	Thermal Resistance	Total	Weight
Capability		1.2	1.5	2	3	7.7	0.3
Ease of Implementation	0.83		1.25	1.67	2.5	6.25	0.25
Cost	0.67	0.8		1.33	2	4.8	0.2
Connectivity	0.5	0.6	0.75		1.5	3.35	0.15
Thermal Resistance	0.33	0.4	0.5	0.67		1.9	0.1
Total						24	1

Figure 21: AHP - Wireless Control

APPENDIX D: Budget & Bill of Materials

Item	Cost	Qty.	Total Cost	Description	URL
1/16 Inch Thick x 12 Inch Wide x 4 Feet Long	\$19.99	2	\$39.98	Neoprene Rubber Sheet - 1/16 Inch Thick x 12 Inch Wide x 4 Feet Long Neoprene Rubber Strips Rolls for DIY Gaskets, Pads, Seals, Crafts, Flooring, Cushioning of Anti-Vibration, Anti-Slip	Amazon.com: Neoprene Rubber Sheet - 1/16 Inch Thick x 12 Inch Wide x 4 Feet Long Neoprene Rubber Strips Rolls for DIY Gaskets, Pads, Seals, Crafts, Flooring, Cushioning of Anti-Vibration, Anti-Slip : Industrial & Scientific
WIFI Arduino(4GB)	\$59	1	\$59	Arduino UNO R3 4-GB RAM [ABX00173] - Hybrid Board, Qualcomm Dragonwing QRB2210 microprocessor (MPU) & STM32U585 Microcontroller(MCU), AI Vision, Voice, IoT, Robotics, Linux Debian OS, Wi-Fi 5, USB-C	https://www.amazon.com/Arduino-4-GB-ABX00173-microprocessor-Microcontroller/dp/B0GFN669S4/ref=sr_1_2_sspa?crid=RNYK7UE58F8F&dib=eyJ2ljoIMSJ9_rhuCFaWKFj8Q2Tb6qTbFYqWGgzlO2GDvSRp0UVdJ-OwXY8do-nVrF_shQS6qXCzSu5zkGWEtDrAXJtYluapFwPYZLNr6PTiFRfHcsuKAKglOzmCJJCGn8v72BQB1KWmRu2qGBG3-VfRwRO8mXfEVXFT1aw4f7qD9XVnH9hZrvXEUS9sOOwXnwdgGFZSlu2-Zj0PFVvWZCQhdiIjGzAzKnNSBijXA0AX2P6rcMsdE.qsyjaySL5SKnPthKrzCdvIZTpvB-FH8e6i7cn4BCi-o&dib_tag=se&keywords=wifi+arduino&qid=1771255258&srefix=wifi+arduino%2Caps%2C126&sr=8-2-spons&sp_csd=d2lkZ2V0TmF1ZT1zcF9hdGY&psc=1
Clevis Pin	\$5	10	\$49	17-4 pH Clevis Pin, 3/16" Diameter, 7/16" Usable Length (93392A109)	https://www.mcmaster.com/products/pins/clevis-pins-5~/clevis-pins-3~/clevis-pins-5~performance-high-strength-2/
Hairpin Cotter Pins	16.38	1	16.38	Zinc-Plated Spring Steel Cotter Pin for 1/16"- 1/8" Pin Diameter, 3/64" Wire Diameter, 1/8" Eye Diameter (98335A027)	https://www.mcmaster.com/products/cotter-pins/hairpin-cotter-pins-1~/

Figure 22: Bill of Materials

Item	Quantity	Cost
20 kg-cm Servo	2	\$ 30
WIFI Arduino(4 GB)	2	\$ 118
Neoprene Rubber Sheet - 1/16" Thick x 12" Wide x 4' Long	1	\$ 39.98
Clevis Pins	1	\$ 5
Hairpin Cotter Pins	1	\$ 16.38
➤ Total Cost = \$ 209.36		

Figure 23: Budget

APPENDIX E: Contact Information

- Christopher Appiah: cappiah1@umbc.edu, 301-787-4126
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- David Inko- Tariah: davidi1@umbc.edu, 240-962-9927
- Joel Matthew: joelm2@umbc.edu, 813-733-9843

APPENDIX F: Compilation of Final Documentation

We have submitted all required documentation. This documentation can be accessed from [this](#) Google Drive under the folder named Team 12: Morphing Wing

- This folder includes:
 - CAD:
 - All solid model files
 - Analysis:
 - FEA files
 - Hand calculations as PDF or Word documents
 - Documentation
 - All Memos
 - The Final Report
 - Presentations
 - All Presentation files

APPENDIX G: Delivery of Prototype and Materials

Our prototype and all related materials are currently in Dr. Yu's lab on the second level of the engineering building in room 237.